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SUPERSEDING
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FEDERAL SPECIFICATION

BLUING, LAUNDRY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of liquid laundry bluing (non-sour, sour) for general laundry use (see 6.1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

CCC-C-430—Cloth, Cotton, Sheeting.
CCC-T-191—Textile Test Methods.
PPP-B-636—Box, Fiberboard.
PPP-C-186—Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The bluing shall consist of a solution of a water-soluble, aniline, blue dye. It shall be free from adulterants and shall not be fast to laundering. When diluted with water in the proportions recommended by the supplier, the bluing shall yield a stock solution which, when applied under the usual conditions of laundry operation, shall tint white goods to a bright white.

FSC 7930

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3.2 **pH.** The pH of the bluing shall be not less than 5.5 and not greater than 8.5 when determined as specified in 4.3.7.

3.3 **Insoluble dye.** There shall be no evidence of insoluble dye when tested as specified in 4.3.7.

3.4 **Presence of aniline blue dye and absence of ultramarine blue, prussian blue and methylene blue.** The bluing, when tested as specified in 4.3.7, shall show the presence of an aniline dye and shall not show the presence of ultramarine blue, prussian blue or methylene blue.

3.5 **Resistance to decomposition by acid and alkali.** A precipitate or sediment shall not form in acid, neutral and alkaline solutions of the bluing when tested as specified in 4.3.7.

3.6 **Fastness to ironing.** Blued fabric shall not fade appreciably nor turn yellow on ironing when tested as specified in 4.3.7.

3.7 **Ease of removal in subsequent washing.** The bluing shall be removed by laundering from cotton fabric when tested as specified in 4.3.7. The blued, washed fabric shall be equal in whiteness to the untreated control fabric.

3.8 **Color intensity of acid, neutral and alkaline bluing solutions.** The bluing, when tested as specified in 4.3.7, shall give transmission values not greater than 30 percent in acid solution, not greater than 45 percent in neutral solution, and not greater than 55 percent in alkaline solution.

3.9 **Difference in the color intensity of acid, neutral, and alkaline bluing solutions.** Based on the percent transmission readings obtained in the color intensity test (see 4.4.8), the alkaline solution shall show a decrease in color intensity (increase in percent transmission) not greater than 15 percent in comparison to the color intensity of the neutral solution. Percent decrease in color intensity shall be calculated as follows:

$$\text{Percent Decrease in Color Intensity} = \frac{T_{alk} - T_n}{T_n} \times 100$$

Where: T_{alk} is percent transmission of alkaline solution.

T_n is percent transmission of neutral solution.

The percent transmission of the acid solution shall not be greater than that of the neutral solution.

3.10 **Directions for use.** Each bottle of laundry bluing shall be durable and legibly marked with the following directions for use:

LIQUID LAUNDRY BLUING
(NON-SOUR, SOUR)

Directions. Prepare a stock solution of the bluing by diluting[1] fluid ounces of the concentrated blue with 5 gallons of fresh water. Rinse the container used to measure the blue with some of the water and add the rinsings to the stock solution. Store the stock solution in glass, enameled, stainless steel, or glazed earthenware containers; do not store in iron or galvanized metal.

Blue is added to the wash-wheel as follows: (1) Use 2 ounces of blue stock solution per 50-pound load; (2) put the proper amount of blue stock solution in a pail and add water to dilute it further; (3) add this pail of blue to the load along the length of the washer while the cylinder, as seen through the supply door, is going down. If the bluing results are too light or too dark, the operator may vary the amount of stock solution used to obtain the desired tint.

3.11 **Workmanship.** The finished product shall be clean and uniform. The quality of the end item shall conform to the levels established in this specification.

[1] Manufacturer shall insert here the number of fluid ounces (to the nearest 1/2 ounce) of his product which is to be used in preparing 5 gallons of stock solution.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. Sampling for inspection shall be in accordance with provisions of MIL-STD-105, except where otherwise indicated.

4.2.1 *Inspection of materials and components.* The suppier is responsible for insuring that materials and components used were tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item and its preparation for delivery shall be examined for the defects in the applicable subparagraphs at the inspection levels and AQLs set forth in 4.3.6. The lot size for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled bottles of the specified capacity for examination in 4.3.2, 4.3.3, and 4.3.4.

4.3.2 *Examination of the primary container for visual defects in construction and markings.* The sample unit for this examination shall be one filled bottle.

Examine	Defect
Construction	Not in container specified. Any leakage of contents. Any split, break, dent, hole, tear, puncture or other defect. Seals or closures not as specified.
Markings (label or printing)	Omitted, illegible, incorrect, incomplete, not in accordance with requirements. Label or printing not clear, smeared, ink not as specified. Directions for use or precautionary markings not as specified (see 3.10).

4.3.3 *Examination for defects in net contents.* The sample unit for this examination shall be one filled bottle. The lot shall be unacceptable if the average net content per bottle is less than the specified or indicated quantity.

4.3.4 *Examination of bluing for defects in workmanship.* The sample unit for this examination shall be the contents of one filled bottle.

Examine	Defect
Workmanship	Not in liquid form. Not clean; not uniform.
Color	Not blue.

4.3.5 *Examination of preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship, contents and weights comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

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Examine	Defect
Markings (exterior and interior	Incorrect; incomplete; illegible; omitted; or improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components; such as incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling. Bulged or distorted containers.
Contents	Number of primary containers per shipping container is more or less than specified.

4.3.6 *Inspection levels and acceptable quality levels (AQLs) for examination.* The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQLS
4.3.2	S-3	2.5
4.3.3	S-2	—
4.3.4	S-2	1.5
4.3.5	S-2	4.0

4.3.7 *Testing of the end item.* The end item shall be tested for the applicable characteristics as indicated in table I for each lot presented. The sample unit shall be one quart of the liquid bluing. The sample size shall be five units. Each sample unit shall be taken from a different shipping container of the end item inspection lot. The test sample shall consist of a two quart composite of the five sample units. A single determination shall be performed on the composite for each characteristic. There shall be no failure of any test to meet the specified requirements.

4.4 Test procedure.

4.4.1 *Stock items.*

4.4.1.1 *Stock solution.* One thousand ml. of stock solution shall be prepared by diluting the bluing with distilled water in the proportions recommended by the manufacturer (see 3.10).

4.4.1.2 *Stock cloth.* Sheeting conforming to CCC-C-430 shall be laundered and shall then be ironed on a flat-bed press as specified in method 5550 of CCC-T-191.

4.4.2 *pH.* The pH of the sample, as received, shall be determined at 25° ± 2°C.,

TABLE I.—*Instructions for testing of the end item*

Characteristic	Specification reference		Number determinations per composite sample	Results reported as	
	Requirement	Test method		Pass or fail	Numerically to nearest
pH	3.2	4.4.2	Average of 2		0.1 pH
Insoluble dye	3.3	4.4.3	1	X	
Presence of aniline blue dye and absence of ultramarine, prussian and methylene blue	3.4	4.4.4	1	X	
Resistance to decomposition by acid and alkali	3.5	4.4.5	1	X	
Fastness to ironing	3.6	4.4.6	1	X	
Ease of removal in washing	3.7	4.4.7	1	X	
Color intensity of acid, neutral, and alkaline bluing solutions	3.8	4.4.8	1	X	
Difference in color intensity of acid, neutral and alkaline bluing solutions	3.9	4.4.8	1	X	

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TABLE II.—*Color comparison*

Reagent	Aniline dye	Ultramarine blue	Prussian blue	Methylene blue
1:1 HCl	Turns slightly darker in color or turns emerald green.	Decolorized	No action	Turns amber
10 percent NaOH	Turns purple or no action	No action	Decolorized	Turns reddish brown
NH_4OH (27 percent NH_3	Turns lighter	No action	Partially decolorized	Turns reddish brown

using a pH meter having a sensitivity and readability of 0.1 pH unit. A sealed type glass electrode shall be used in conjunction with a potassium chloride-calomel reference electrode. The pH meter shall be calibrated before use against a standard buffer solution of pH 7.

4.4.3 *Insoluble dye.* The sample, as received, shall be thoroughly mixed by shaking and 10 ml. of the solution shall be added by means of pipette, into a 1,500-ml. breaker, shall be diluted with 1,000 ml. of distilled water, and shall be stirred well. The solution shall be filtered through a Whatman No. 12 filter paper (32.0 cm.). The filter paper shall be visually observed for deposits of insoluble matter. (Readily and completely soluble dyes color the paper uniformly.)

4.4.4 *Presence of aniline blue dye and absence of ultramarine, prussian and methylene blue.* Thirty-five ml. of distilled water shall be added to each of four 75-ml. test tubes containing 5 ml. of stock solution. One of these test tubes shall be retained as a control. To the others shall be added 10 ml. of 1:1 hydrochloric acid, 10 ml. of 10 percent sodium hydroxide, and 10 ml. of concentrated ammonium hydroxide, respectively. The solution shall be mixed well and the color changes shall be compared with table II.

4.4.5 *Resistance to decomposition by acid and alkali.* Two ml. of stock solution shall be added, by means of pipette, to each of three 250 ml. beakers containing 100 ml. of buffer solution having a pH of 4.0, 7.0 and 8.0 respectively. The buffer solution shall be prepared as follows:

pH	Buffer solution composition
4.0	0.40 ml. 0.1*N* NaOH + 50 ml. 0.1M $KHC_8H_4O_4$ + distilled water to 100 ml.
7.0	29.63 ml. 0.1*N* NaOH + 50 ml. 0.1M KH_2PO_4 + distilled water to 100 ml.
8.0	3.97 ml. 0.1*N* NaOH + 50 ml. 0.1M H_3BO_3 + distilled water to 100 ml.
0.1N	Sodium hydroxide (NaOH) prepared carbonate free and standardized according to usual volumetric procedure.
0.1M	Potassium acid phthalate ($KHC_8H_4O_4$) — molecular weight 204.2 - ACS reagent grade is of sufficient purity.
0.1M	Monopotassium phosphate (KH_2PO_4) - molecular weight 136.2 - ACS reagent grade shall be recrystallized twice from water and dried in oven at 110° - 120°C.
0.1M	Boric acid (H_3BO_3) - molecular weight 62.0 - ACS reagent grade shall be recrystallized and dried in air.

The buffer solution with bluing shall be stirred well and shall stand for 24 hours. The solution shall be observed for the presence of precipitation or sediments.

4.4.6 *Fastness to ironing.* A specimen of stock cloth approximately 8 by 8 inches in dimension shall be prepared and shall be soaked for 5 minutes, at 90° + 5°F., in 1,000 ml. of distilled water to which has been added 15 ml. of stock solution. The cloth shall be removed and shall be passed once through the wringer of a washing machine (do not wring by hand). The cloth shall be dried in air. The specimen shall be cut in

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half and one half shall be ironed as specified in method 5550 of CCC-T-191. Using the other half as a control, color comparisons shall be made by visual observation to determine whether the sample has faded appreciably or turned yellow.

4.4.7 *Ease of removal in subsequent washing.* Two specimens of stock cloth about 2-1/2 by 4 inches in dimensions shall be cut. One specimen shall be soaked for 5 minutes, at 90° ± 5°F. in 1,000 ml. of distilled water to which has been added 15 ml. of stock solution. The cloth shall be dried in air. The other specimen of cloth shall be used as a control. Both specimens shall be washed separately, as specified in method 5610 of CCC-T-191. The bluing shall be removed completely from the sample cloth as determined by comparison with the control specimen.

4.4.8 *Color intensity of acid, neutral, and alkaline bluing solution.*

4.4.8.1 *Apparatus.*

Coleman Universal Spectrophotometer or equal. The spectrophotometer shall be used with a 550 millimicron wave length setting, a PC-4 glass filter, and a 13 by 13 by 105 millimeter (mm.) square cuvette. It shall be adjusted to read 100 on the galvonometer using distilled water.

4.4.8.2 *Procedure.* Three hundred ml. of a pH 4.0 buffer solution (see 4.4.5) shall be added to a 400-ml. beaker. By means of a pipette or burette, exactly 5 ml. of stock shall be added to the beaker, stirred well, and allowed to stand for exactly 5 minutes. At the end of 5 minutes, the color intensity of the solution shall be measured with the previously standardized Coleman instrument. The measurement shall be repeated using buffer solutions of pH 7.0 and 8.0.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* One quart or one gallon of bluing shall be packaged in a plastic bottle conforming to group A, class 2, style 1, closure A, seal A of PPP-C-186.

5.1.2 *Level C.* Bluing shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Twenty-four 1-quart bottles or six 1-gallon bottles of bluing, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to class 2, V3c or V3s, style RSC-L of PPP-B-636. Each shipping container shall be provided with a taped, one-piece, double walled, corrugated fiberboard liner, and top and bottom pads conforming to class 1 requirements of PPP-B-636. The minimum bursting strength of the fiberboard for the liner and pads shall be 275 p.s.i. Top and bottom pads shall be cut the full inside length and width of the fiberboard container. The height of the liner shall be equal to the inside depth of the container less the thickness of the top and bottom pads. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 *Level B.* Twenty-four 1-quart bottles or six 1-gallon bottles of bluing packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to type I or II, class 1, style RSC-L of PPP-B-636. Each fiberboard shipping container shall be provided with a liner and top and bottom pads as specified in 5.2.1. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

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5.2.3 *Level C.* Bluing, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 *Labeling.* Each bottle of liquid bluing shall be durably and legibly marked with the instructions for use as specified in 3.10.

6. NOTES

6.1 **Intended use.** The laundry bluing covered by this specification is intended for general laundry use to neutralize the natural slightly yellow shade of fabrics by coloring them with a faint blue tint. The bluing works satisfactorily under acid, alkaline, or neutral conditions and may therefore be used whether or not souring is done.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1).

6.3 **Basis for award.** The basis for award should be determined by the following formula:

$$\text{(Delivery price per quart)} \times \frac{A}{5}$$

Where A = number of fluid ounces of bluing recommended to prepare 5 gallons of stock solution (see 3.10).

6.4 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Bluing, laundry, liquid.
Carload minimum weight 36,000 pounds.

Motor:

Bluing, laundry, not otherwise indexed, liquid.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—SH

Air Force—69

Interested activities:

Review

Army—GL, MD, SM

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Navy—SH, MC

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Army GL

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